

Kotaro Ikeda

📍 Tokyo, Japan ✉ kotatumuri114@icloud.com 🌐 kotatumuri-room.github.io

About

My name is Kotaro Ikeda. I am interested in machine learning as well as its intersections with thermodynamics and dynamical systems.

Education

- The University of Tokyo**, Junior Division, College of Arts and Sciences – Apr 2022 – Mar 2024
Tokyo, Japan
- BE The University of Tokyo**, Department of Mathematical Engineering and Apr 2024 – present
Information Physics, School of Engineering – Tokyo, Japan
- Planned to graduate in March 2026

Publications

- Pairwise Optimal Transports for Training All-to-All Flow-Based Condition Transfer Model** Dec 2025
Kotaro Ikeda, Masanori Koyama, Jinzhe Zhang, Kohei Hayashi, Kenji Fukumizu
NeurIPS 2025 Poster (Acceptance rate: 24.52%)
[10.48550/arXiv.2504.03188](https://arxiv.org/abs/10.48550/arXiv.2504.03188)
- Speed-accuracy relations for diffusion models: Wisdom from nonequilibrium thermodynamics and optimal transport** June 2025
Kotaro Ikeda, Tomoya Uda, Daisuke Okanohara, Sosuke Ito
Physical Review X 15:3 (2025): 031031 (Top Journal in Physics [Impact Factor: 15.7])
[10.1103/x5vj-8jq9](https://arxiv.org/abs/10.1103/x5vj-8jq9)

Projects

- Research Experience Seminar on the Universal Principles of Life** Apr 2023 – Mar 2024
Small-group seminar by Universal Biology Institute, the University of Tokyo
- Constructed a novel analogy between diffusion models and stochastic thermodynamics
 - Theoretically discussed the optimality of noise schedules in diffusion models
 - Published a paper on this topic in a top-physics journal
 - Press Release is [here](#)
- Board Game agent development@May Fes** Feb 2025 – May 2025
Created a reinforcement learning based agent for Blokus™
- State-of-the-art performance against publically available agents.
 - Created a website and a real-world competition system for playing with our agents.
 - Project done for [May Fes](#) with my classmates.
 - Served as a project leader.

Fundings

- Funai Overseas Scholarship** Sept 2026 – Aug 2028
Partial funding for studying abroad.
- \$36,000 per year for living cost
 - \$14,000 per year for tuition

- full coverage of medical insurance

Experience

Summer Intern, Preferred Networks, Inc. – Tokyo, Japan

Aug 2024 – Feb 2025

- Developed deep learning models for style transfer based on flow matching and optimal transport theories.
- Preprint of this work can be found in the publications section.

Part-time Engineer, Akari, Inc. – Tokyo, Japan

June 2023 – Mar 2024

- Enrolled in a project involving NeRF and 3D Gaussian splatting.

Part-time Engineer, Meltly, Inc. – Tokyo, Japan

Apr 2025 – present

- Worked for training a generative model for anime-character images.

Skills

Programming: Proficient with Python, PyTorch and Git; good understanding of C++, Web development (React) and App development (Swift, Java)

Languages: English (fluent, TOEFL: 105/120), Japanese (native)

Conferences in Japan

Speed-accuracy trade-off for the diffusion models: Wisdom from nonequilibrium thermodynamics and optimal transport

Kotaro Ikeda (presenter), Tomoya Uda, Daisuke Okanohara, Sosuke Ito

GeoBeta Flow Matching

Nov 2024

Kotaro Ikeda (presenter), Masanori Koyama, Jinzhe Zhang, Kohei Hayashi, Kenji Fukumizu

The 27th Information-Biased Induction Sciences Workshop (IBIS), Saitama, Japan